UNIVERSAL GATES

LAB 4

CSE-202L DIGITAL LOGIC GATES

Submitted by: SYEDA AFIA SHAH, MAIRA ZAFAR

Reg No.: 2115, 2106

Group 4

Section C

Submitted to Dr. Rehmat Ullah Khattak

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Department of Computer Systems Engineering

University of Engineering and Technology, Peshawar

OBJECTIVES:

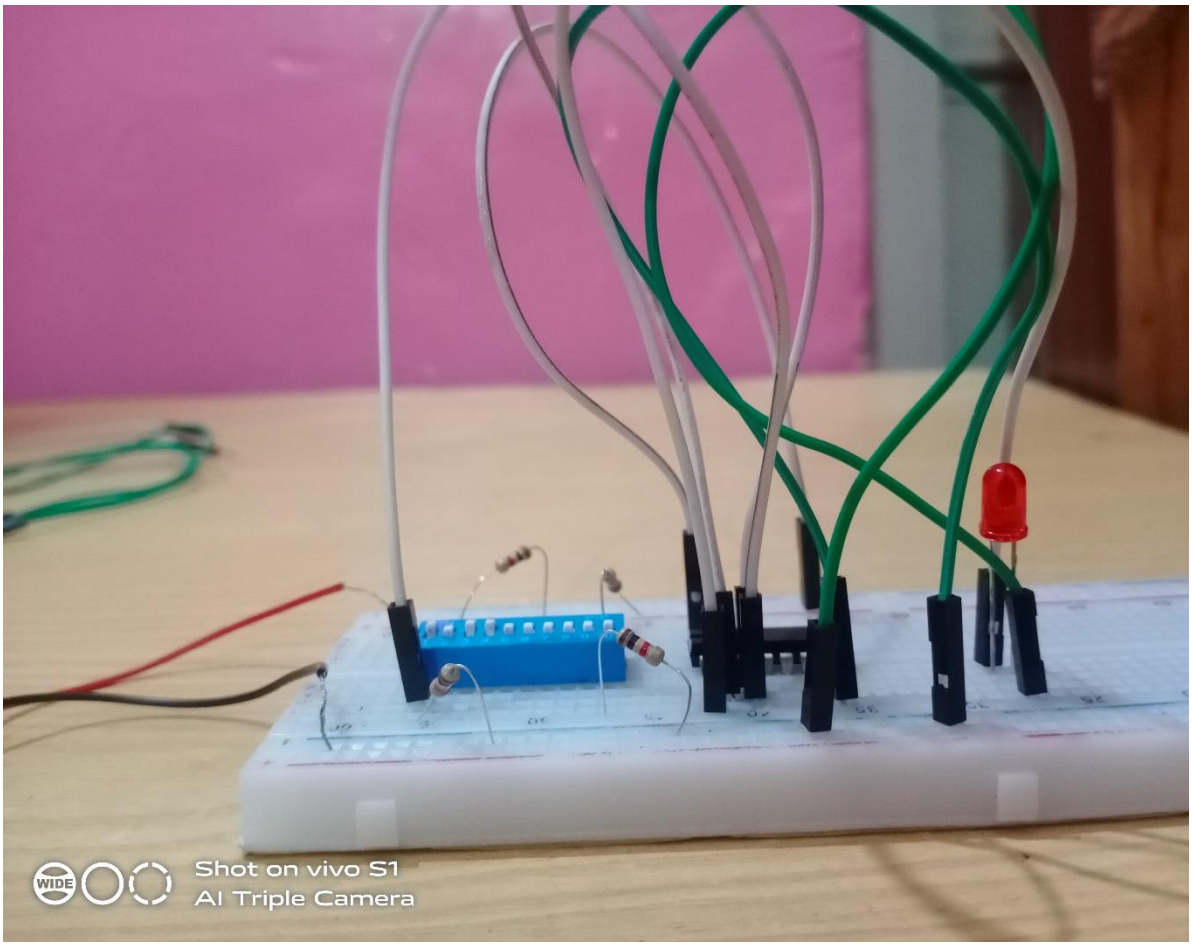
- To use universal gates to implement basic gates.
- Implementing complex circuits using Universal Gates.

COMPONENTS:

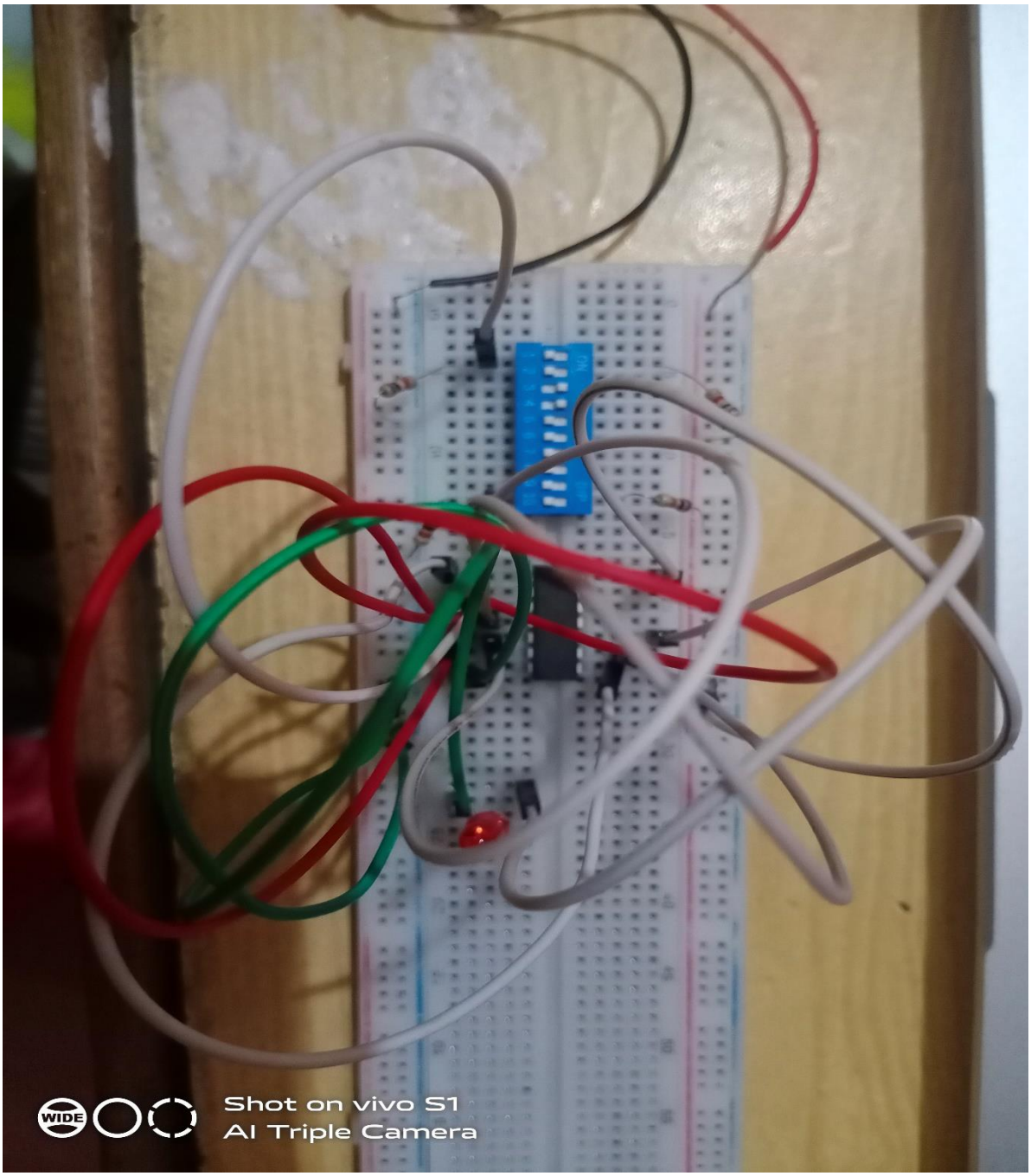
- 7400 and 7402 IC's
- Jumper wires
- 9V battery
- 1kΩ resistors
- Dip switch

MANIFESTING BASIC GATES USING NAND GATES

USING IC 7400



NOT



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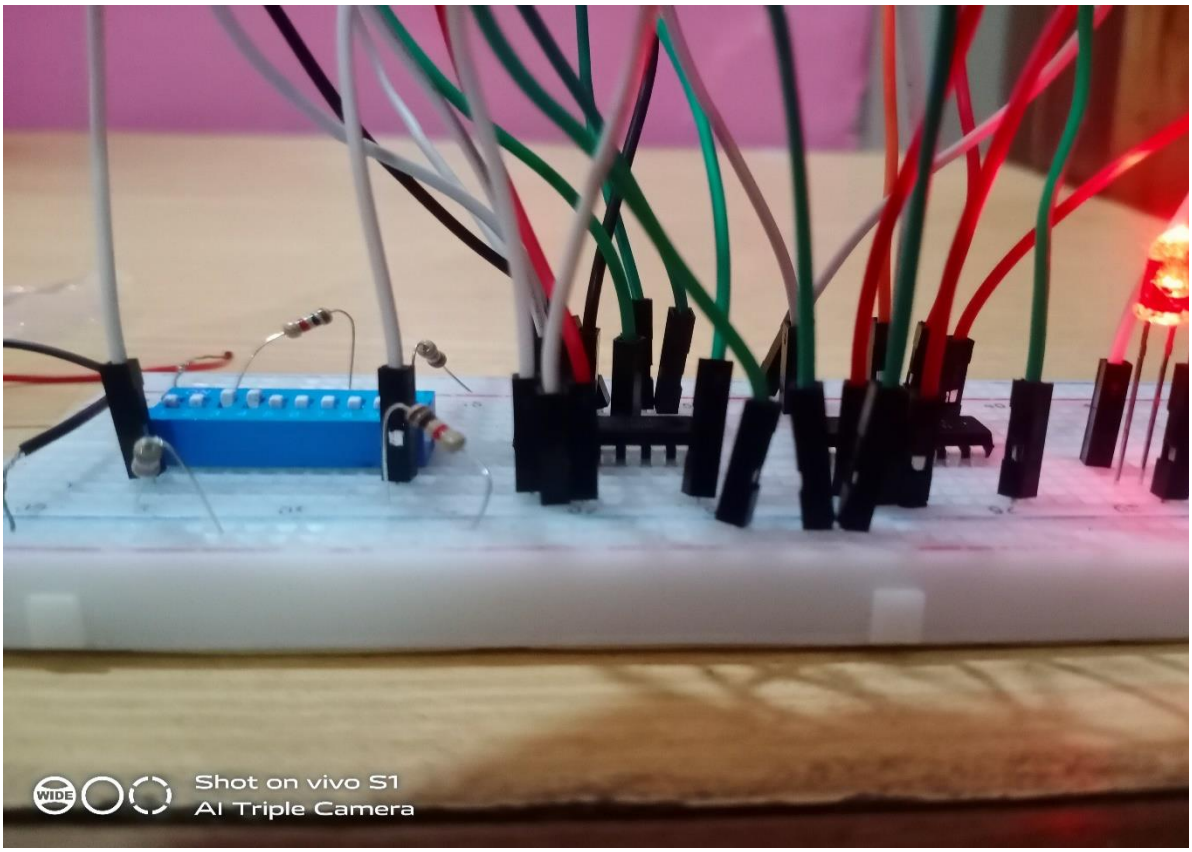
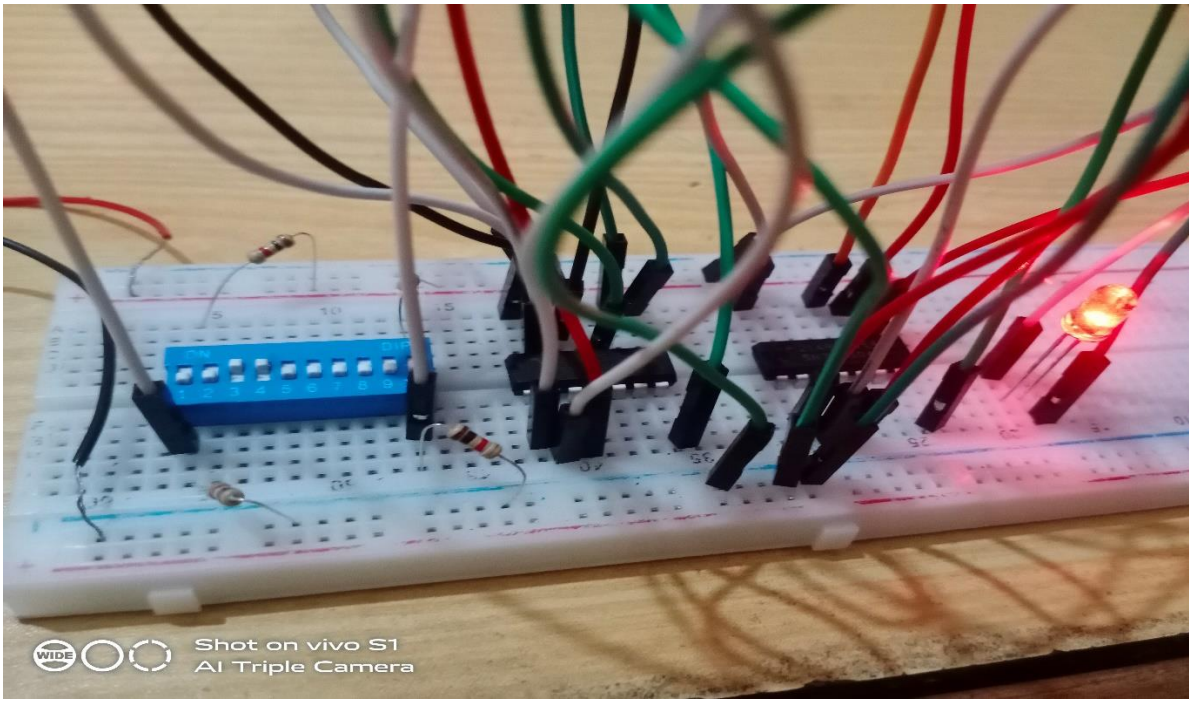

Shot on vivo S1
AI Triple Camera

XOR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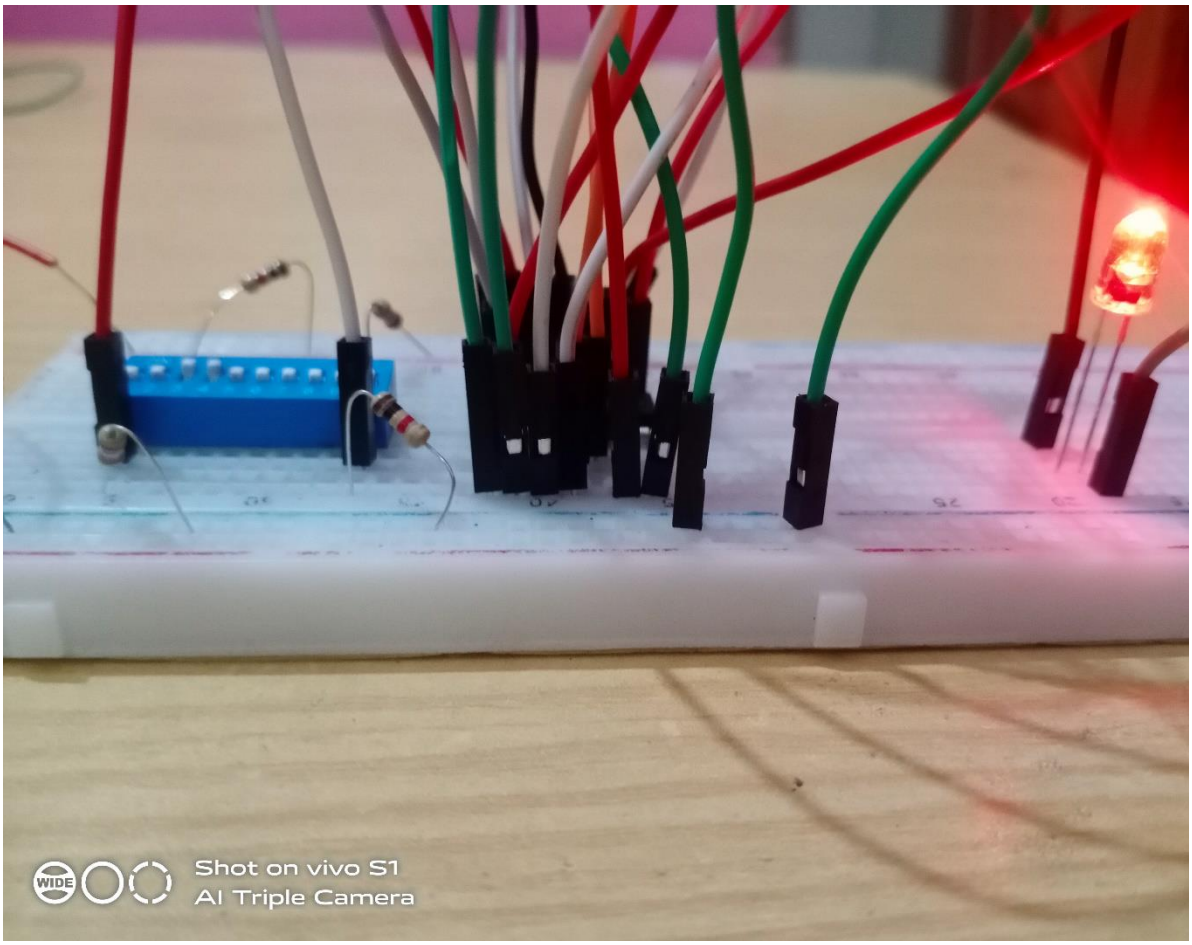
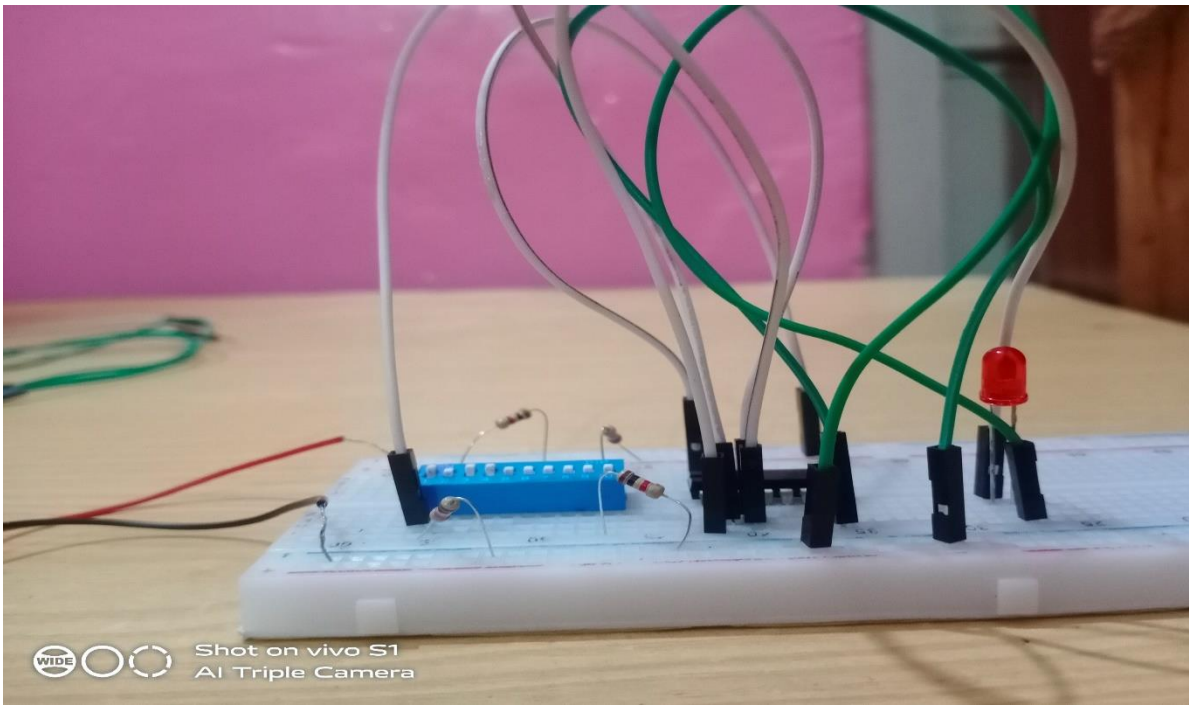
XNOR



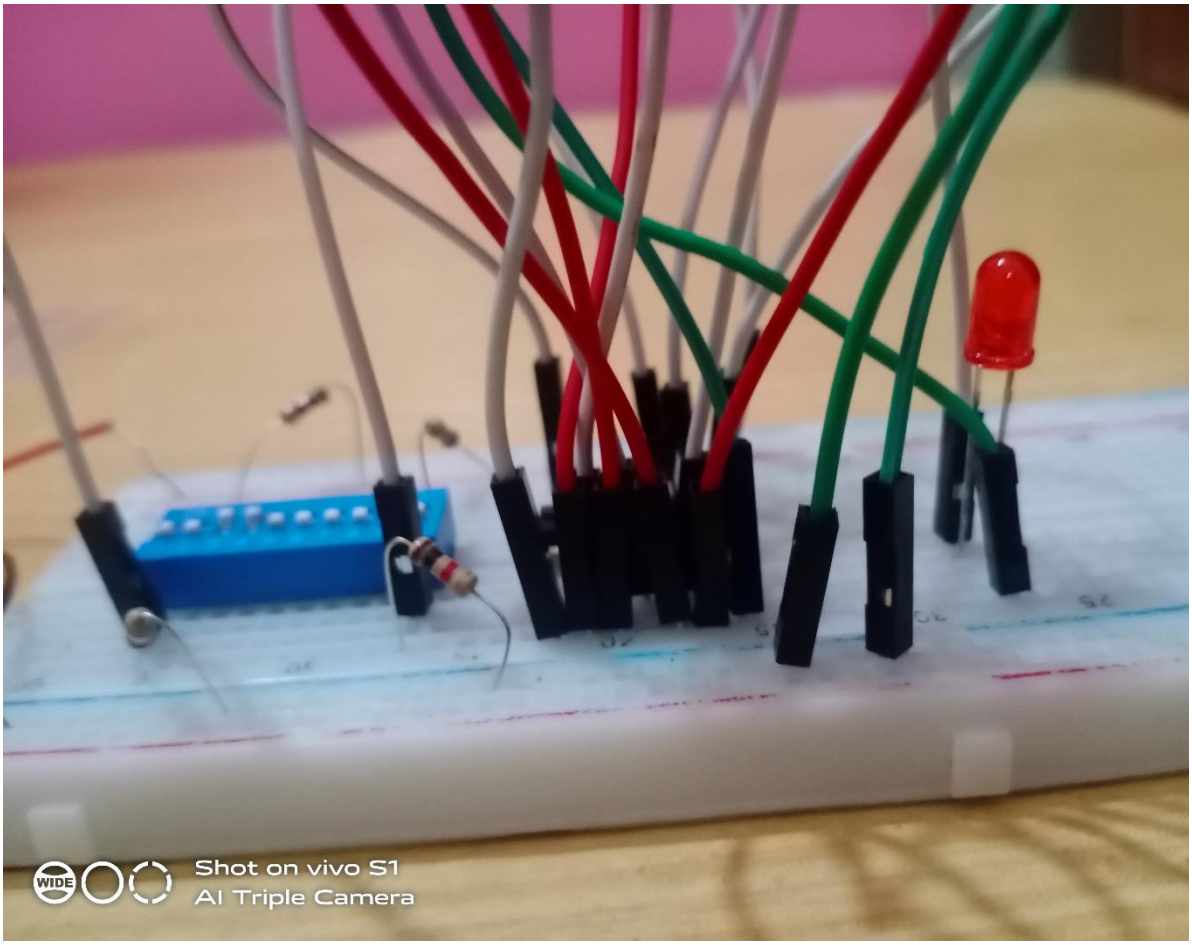
NOR

MANIFESTING BASIC GATES USING NOR GATES

USING IC 7402


Shot on vivo S1
AI Triple Camera

NOT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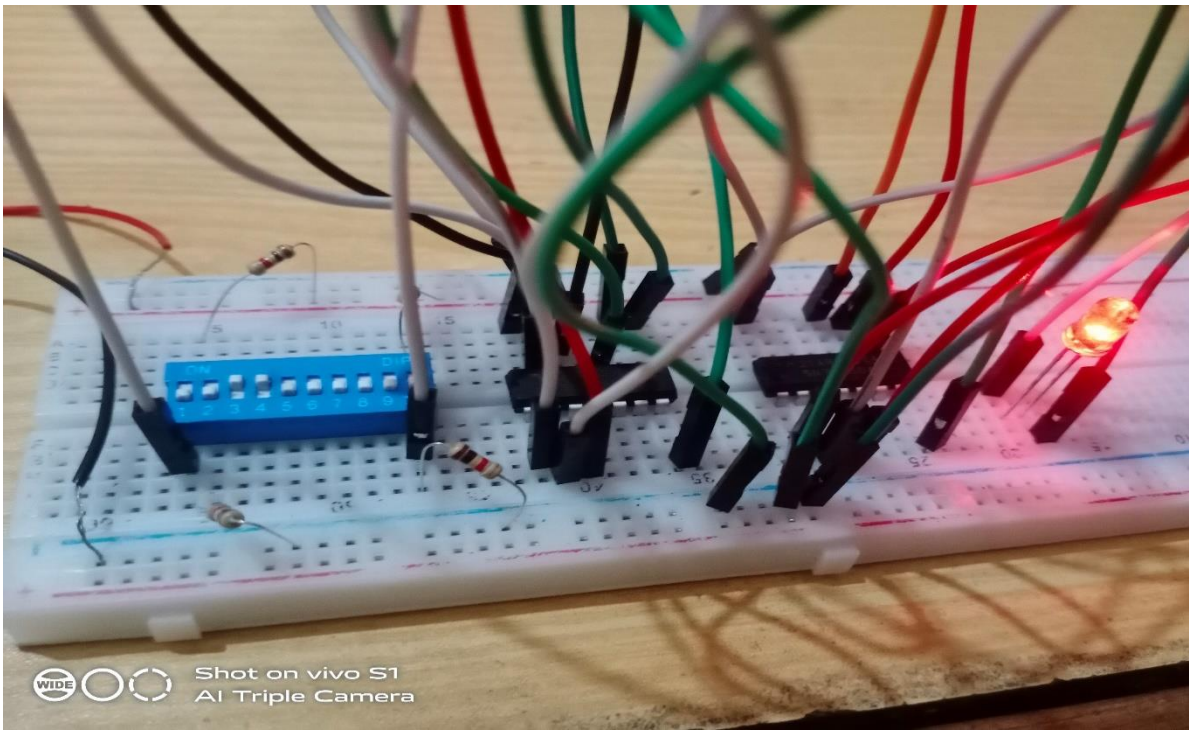
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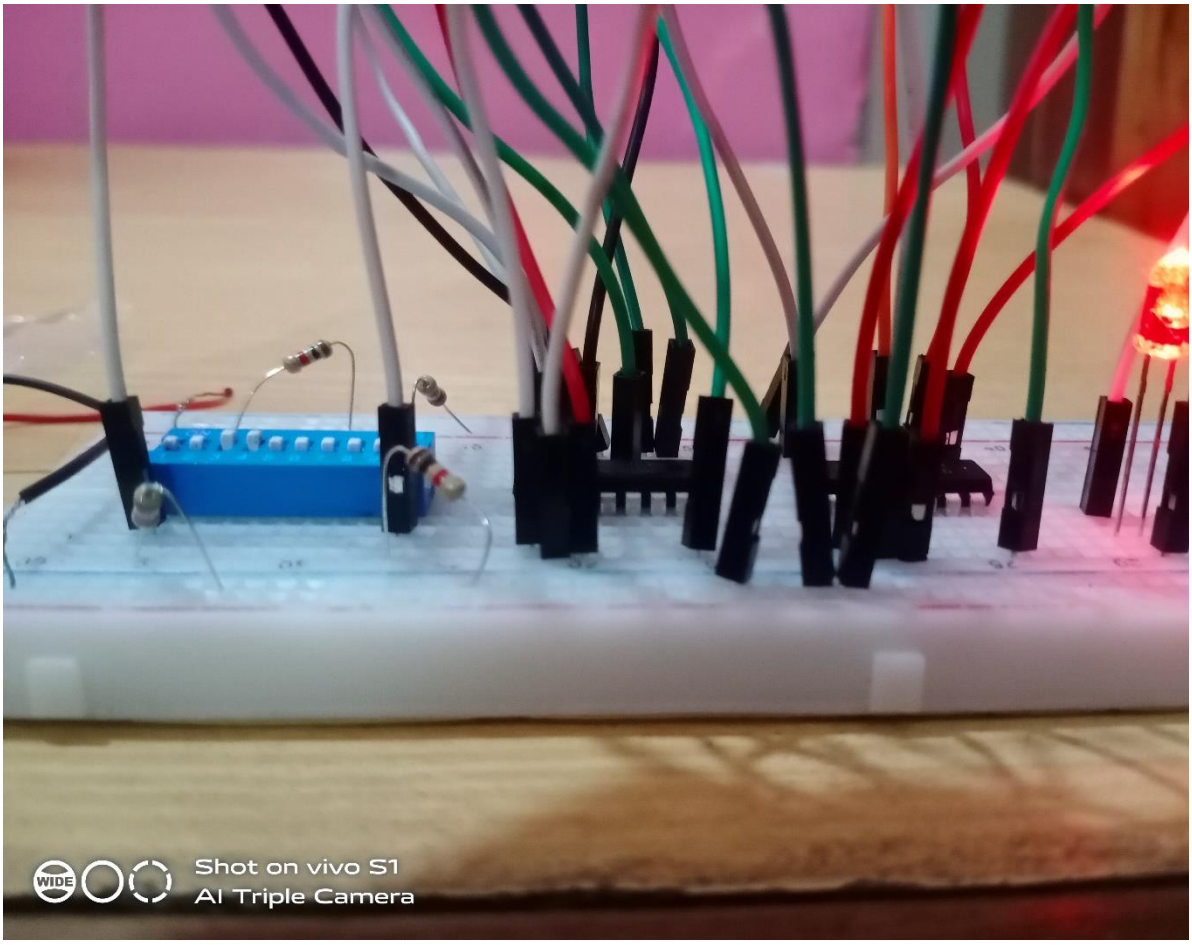
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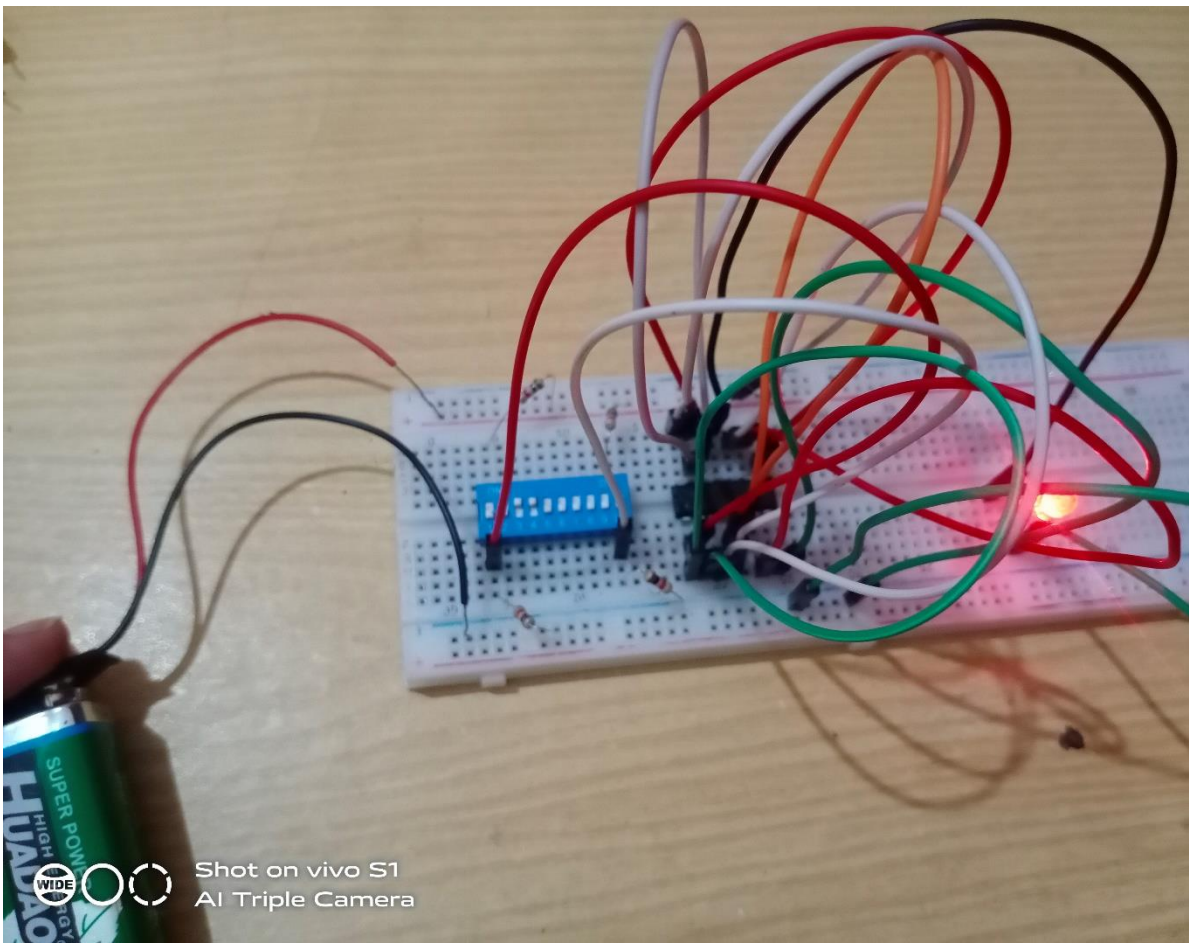


NAND

TASK 1

- Derive function for the given Truth Table.
- Manifest the derived function first using NAND gates only.

A	B	C	F(output)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Deriving Function for the given truth table:

MINTERM: The term with product of N literals occurring exactly once.

MAXTERM: The term with the sum of N literals occurring exactly once.

MATHEMATICAL DERIVATION:

We'll derive the function using minterms.

$$F = m1 + m3 + m6 + m7$$

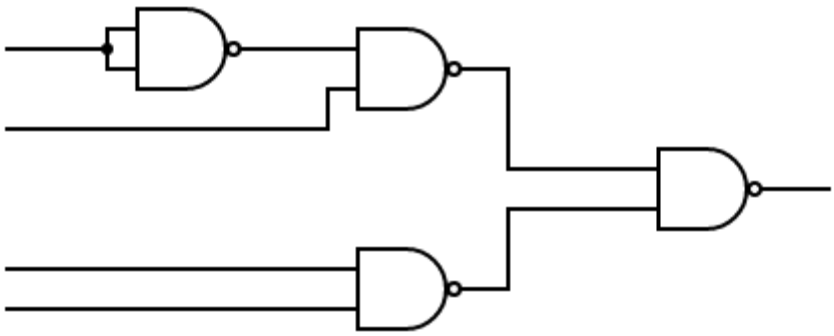
$$F = A'B'C + A'BC + ABC' + ABC$$

$$F = A'C(B' + B) + AB(C' + C)$$

$$F = A'C + AB$$

MANIFESTING FUNCTION USING NAND GATES:

DIGITAL CIRCUIT:



PROCEDURE:

- Check switches of the DIP switch and all gates of the 7400 IC.
- 7400 has 4 NAND gates. We'll use all 4. Power pin 14, and ground pin 7.
- To get A', Give input to gate (1) of the IC from switch 2. Short the two inputs of the gate by connecting both columns of the breadboard via jumper wire.
- Take the output A' and pass it to gate (2) as input.
- Take C from switch 8 as input for gate (2). Pass the output $\overline{(\bar{A}.C)}$ to gate 4.
- Take input A from switch 2, and B from switch 4. Pass the output $\overline{A.B}$ to gate 4.
- The two inputs of gate 4; that are $\overline{(\bar{A}.C)}$ and $A.B$ will return $(\bar{A}.C) + A.B$
- Pass the output of gate 4 to LED.

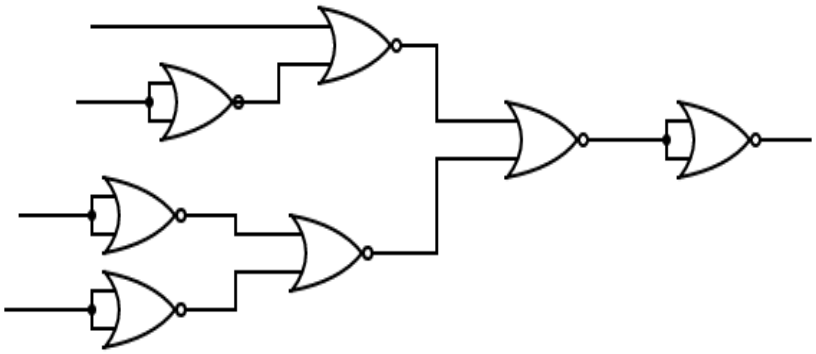
TRUTH TABLE DRAWN ON EXPERIMENTAL DATA:

A	B	C	F(output)
L	L	L	L
L	L	H	H
L	H	L	L
L	H	H	H
H	L	L	L
H	L	H	L
H	H	L	H
H	H	H	H

TASK 2

MANIFESTING FUNCTION USING NOR GATES:

DIGITAL CIRCUIT:



PROCEDURE:

- Take two 7402 IC's. We'll use seven gates in total to manifest the derived function.
- Switches used are: 2,4,8. Switch 2 is input A, 4 is input B, and 8 is input C.
- Pass input C to gate 1 of the IC. Short the two input columns via jumper wire. Pass the output C' to gate 4.
- Pass input A to gate 4.
- Pass the output of gate 4, that is (A+C')' to gate 6.
- Take input A and B, and pass them to gate 2, and 3.
- Take output from gate 2, that is A' and pass to gate 5.
- Take output from gate 3, that is C' and pass to gate 5.
- Now, pass the output of gate 5 that is (A'+B')' to gate 6.

- Finally, pass the output of gate 6, that is ((A'C)+(AB))' to gate 7. Short the two input columns to invert the output.
- Pass the inverted output from gate 7 to LED.

TRUTH TABLE DRAWN ON EXPERIMENTAL DATA:

A	B	C	F(output)
L	L	L	L
L	L	H	H
L	H	L	L
L	H	H	H
H	L	L	L
H	L	H	L
H	H	L	H
H	H	H	H

CONCLUSION:

- We can manifest any circuit using Universal gates.
- NAND gates are faster than other gates. Therefore, they are used commercially in digital circuits.